

Lab 04: Selection Statements II

Objective

The purpose of this lab is to familiarize students with Conditional statements and explain how to control the sequence of statement execution, depending on the value of an expression using the decision making statements (if-then, if-then-else, switch). There are some exercises, through which they will understand the concept learned in this chapter.

Current Lab Learning Outcomes (LLO)

Students will learn how to write Java programs that make decisions, and execute different parts depending on a Boolean true/false value. Students will solve different problems that will help them to choose appropriate selection control statement (single if, block if, multiple if or switch) for a given task. There are some exercises, through which they will understand the concept learned in this chapter.

Lab Requirements

NetBeans

Lab Description

Selection Statements

Java has several types of selection statements:

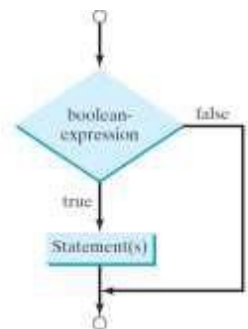
– one-way if statements, two-way if-else statements, nested if statements, multi-way if-else statements, switch statements, and conditional expressions

One-way if Statements

A one-way if statement executes an action if and only if the condition is true.

Syntax:

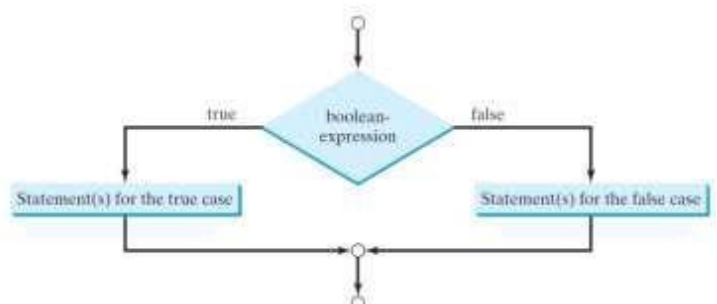
```
if (boolean-expression) {  
    statement(s);  
}
```



Two-way if-else Statements

Syntax:

```
if (boolean-expression) {  
    statement(s)-for-the-true-case;  
}  
else {  
    statement(s)-for-the-false-case;  
}
```



If the **boolean-expression** evaluates to true, the **statement(s) for the true case** are executed. Else, the **statement(s) for the false case** are executed.

Nested if and Multi-Way if-else Statements

- The statement inside an if or if-else statement can be any legal Java statement including another if or if-else statement!
- The inner if statement is said to be nested inside the outer if statement.

Switch Statement

- Java provides a switch statement to simplify coding for multiple conditions.

Syntax for the switch statement

```
switch (switch-expression)
{
    case value1: statement(s)1;
                break;
    case value2: statement(s)2;
                break;
    ...
    case valueN: statement(s)N;
                break;
    default: statement(s)-for-default;
}
```

Practice Activity with Lab Instructor (10 minutes)

Write a program that prompts the user to enter a number, and displays **This number is positive** if the entered number is positive otherwise, displays **This number is negative** if it is negative . Here is a sample run:

```
Enter integer number: 10
This number is positive
```

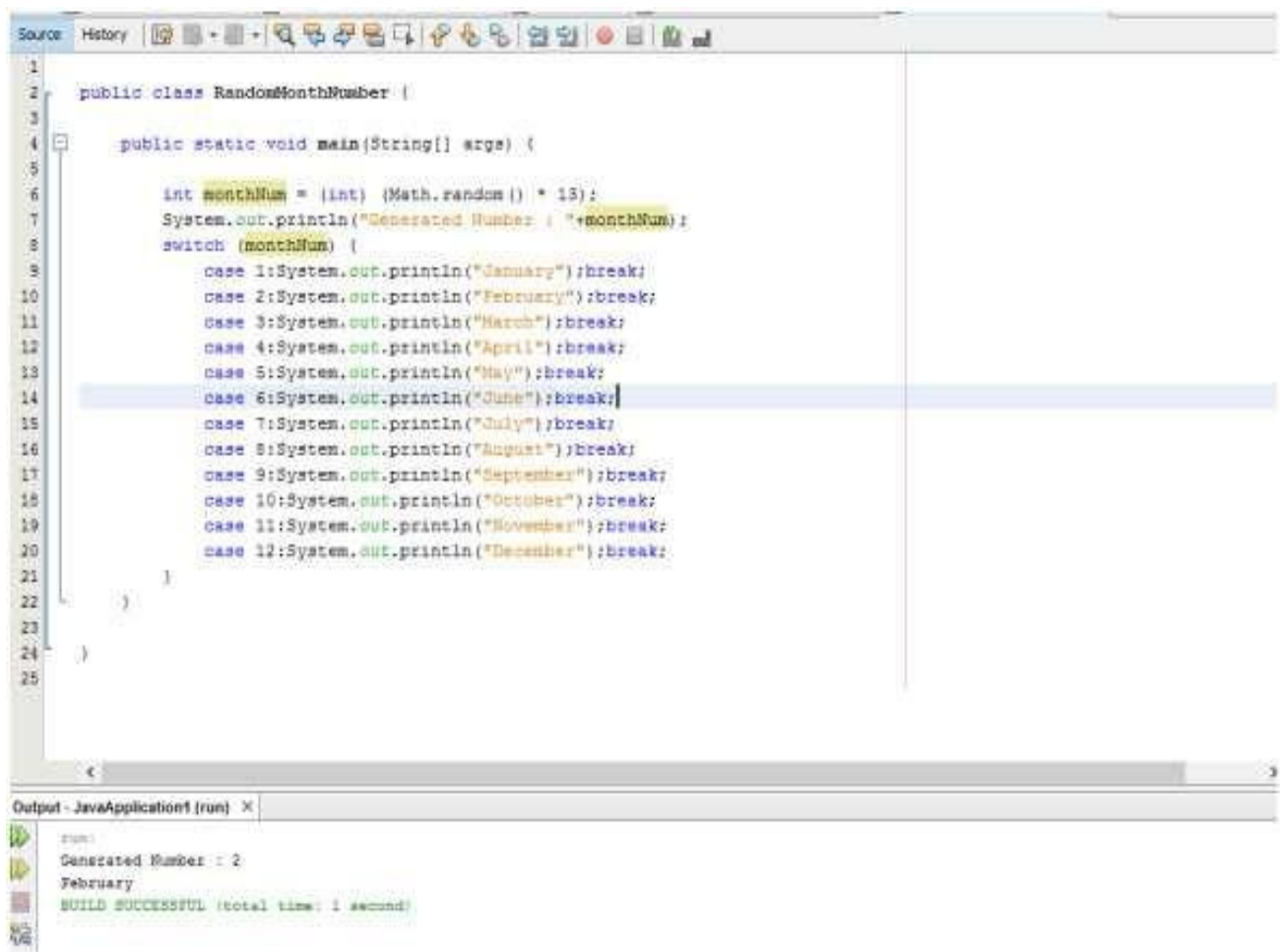
```
Enter integer number: -5
This number is negative
```

Input: Value of variable *number*.

Processing and Output: Check the *number*, if it is greater than zero display the first message, else display the second message.

Solution:

1. Open NetBeans and create a new project
2. Create a new java main class and write its name as **RandomMonthNumber**
3. Write the following code inside the main method



```
1 public class RandomMonthNumber {
2
3
4     public static void main(String[] args) {
5
6         int monthNum = (int) (Math.random() * 13);
7         System.out.println("Generated Number : "+monthNum);
8         switch (monthNum) {
9             case 1: System.out.println("January"); break;
10            case 2: System.out.println("February"); break;
11            case 3: System.out.println("March"); break;
12            case 4: System.out.println("April"); break;
13            case 5: System.out.println("May"); break;
14            case 6: System.out.println("June"); break;
15            case 7: System.out.println("July"); break;
16            case 8: System.out.println("August"); break;
17            case 9: System.out.println("September"); break;
18            case 10: System.out.println("October"); break;
19            case 11: System.out.println("November"); break;
20            case 12: System.out.println("December"); break;
21        }
22    }
23 }
24
25
```

Output - JavaApplication1 [run] X

```
run:
Generated Number : 2
February
BUILD SUCCESSFUL (total time: 1 second)
```

- Short Exercise (5 minutes)

Rewrite the above program using **if** statement.

Lab Assessment

Three Problems\ exercises should be completed **in 60 minutes individually**

1. Write a program that prompts the user to enter a number and prints "ONE" if the user enter 1, "TWO" if the user enter 2 ,.....till "NINE" or prints "OTHER" if the user enter number not in 0,1,2,...9.
2. (**Game: scissor, rock, paper**) Write a program that plays the popular scissor-rock paper game. (A scissor can cut a paper, a rock can knock a scissor, and a paper can wrap a rock.) The program randomly generates a number 0, 1, or 2 representing scissor, rock, and paper. The program prompts the user to enter a number 0, 1, or 2 and displays a message indicating whether the user or the computer wins, loses, or draws. Here are sample runs

```
scissor (0), rock (1), paper (2): 1 Enter
The computer is scissor. You are rock. You won
```

```
scissor (0), rock (1), paper (2): 2 Enter
The computer is paper. You are paper too. It is a draw
```

3. (**Cost of shipping**) A shipping company uses the following function to calculate the cost (in dollars) of shipping based on the weight of the package (in pounds).

$$c(w) = \begin{cases} 3.5, & \text{if } 0 < w \leq 1 \\ 5.5, & \text{if } 1 < w \leq 3 \\ 8.5, & \text{if } 3 < w \leq 10 \\ 10.5, & \text{if } 10 < w \leq 20 \end{cases}$$

Write a program that prompts the user to enter the weight of the package and display the shipping cost. If the weight is greater than 50, display a message "the package cannot be shipped."

4. (**Financials: currency exchange**) Write a program that prompts the user to enter the exchange rate from currency in U.S. dollars to Saudi SAR. Prompt the user to enter 0 to convert from U.S. dollars to Saudi SAR and 1 to convert from Saudi SAR and U.S. dollars. Prompt the user to enter the amount in U.S. dollars or Saudi SAR to convert it to Saudi SAR or U.S. dollars, respectively. (**Dollars to Riyals, with an exchange rate of 3.75**)